

Automated Customer Reviews Analysis using NL



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Project Overview

•Goal: To develop an NLP-powered system that automates the analysis of Amazon product reviews.

Key Tasks:

- 1.Sentiment classification
- 2.Product category clustering
- 3.Generative review summarization



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CLUSTERING

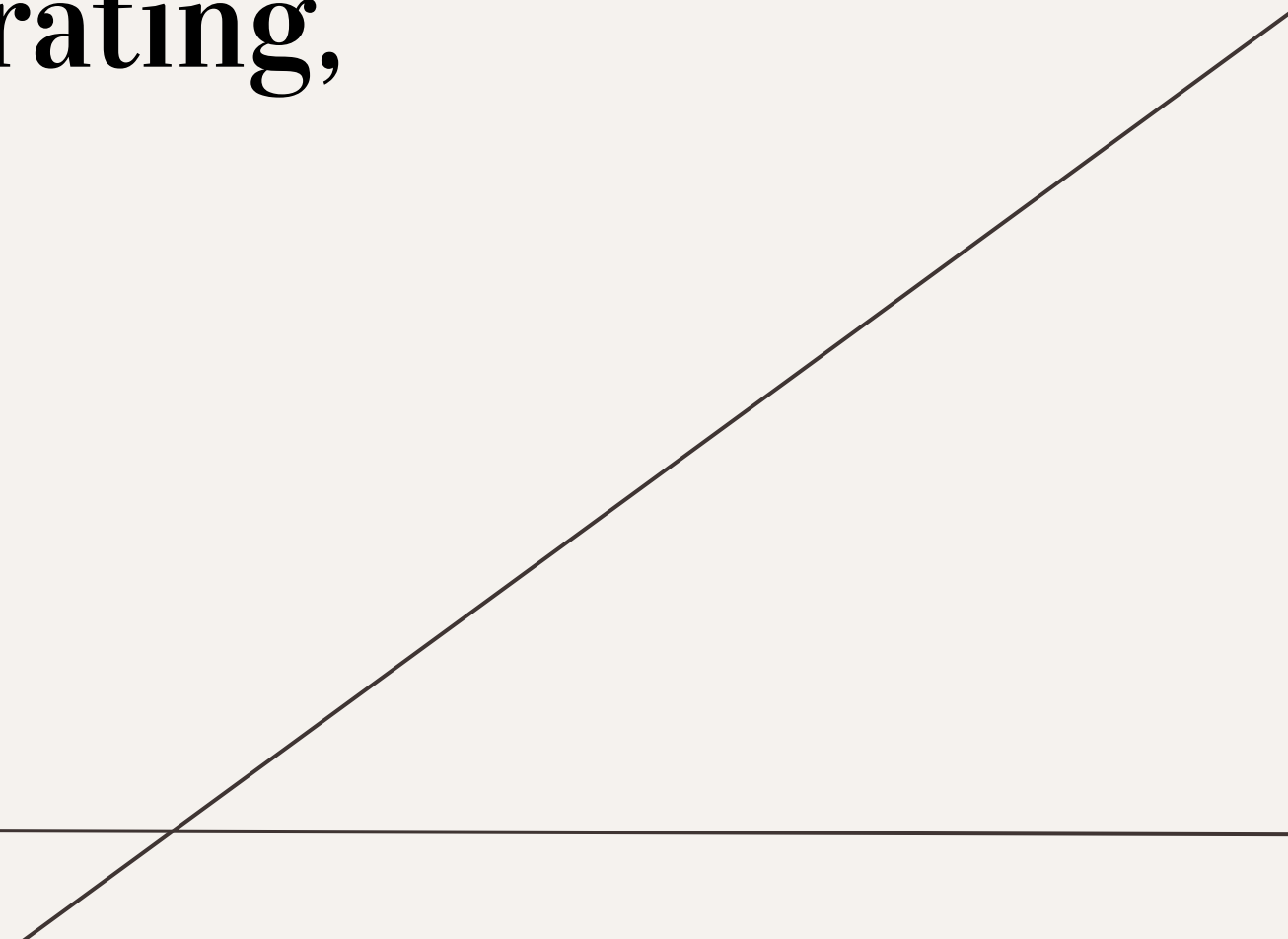
03

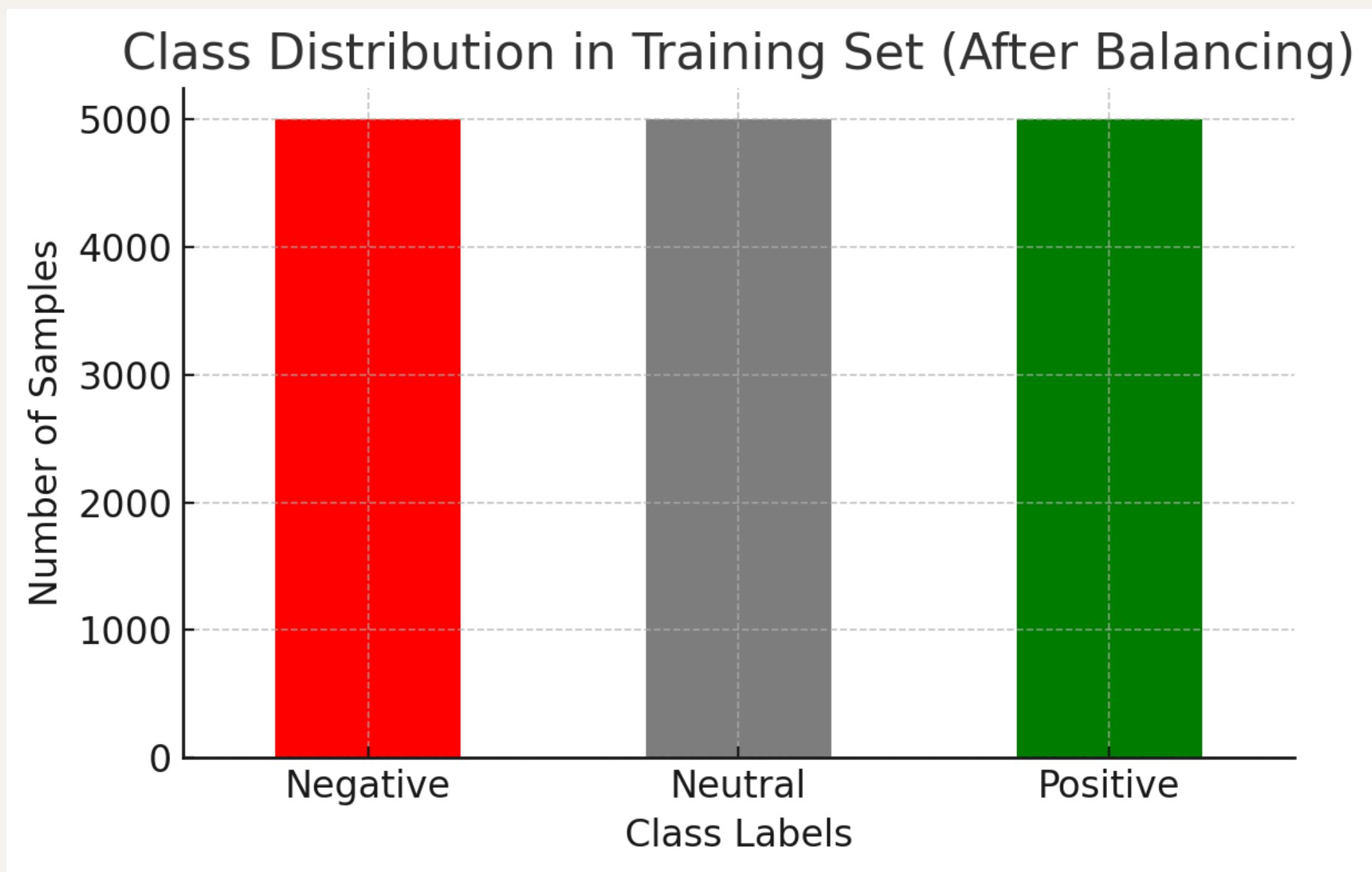
SUMMARIZATION

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Conclusion

DATASET OVERVIEW

- Amazon Consumer Reviews of_Amazon Products
 - 28,000+ reviews (May 2019)
 - Products: Tablet, Batteries, Accessories, etc.
 - Key columns used: reviews.text, reviews.rating, primaryCategories, name, categories.
- 
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01

CLASSIFICATION

Data Preprocessing

CLASSIFY REVIEWS INTO POSITIVE, NEUTRAL, OR NEGATIVE CATEGORIES USING NLP

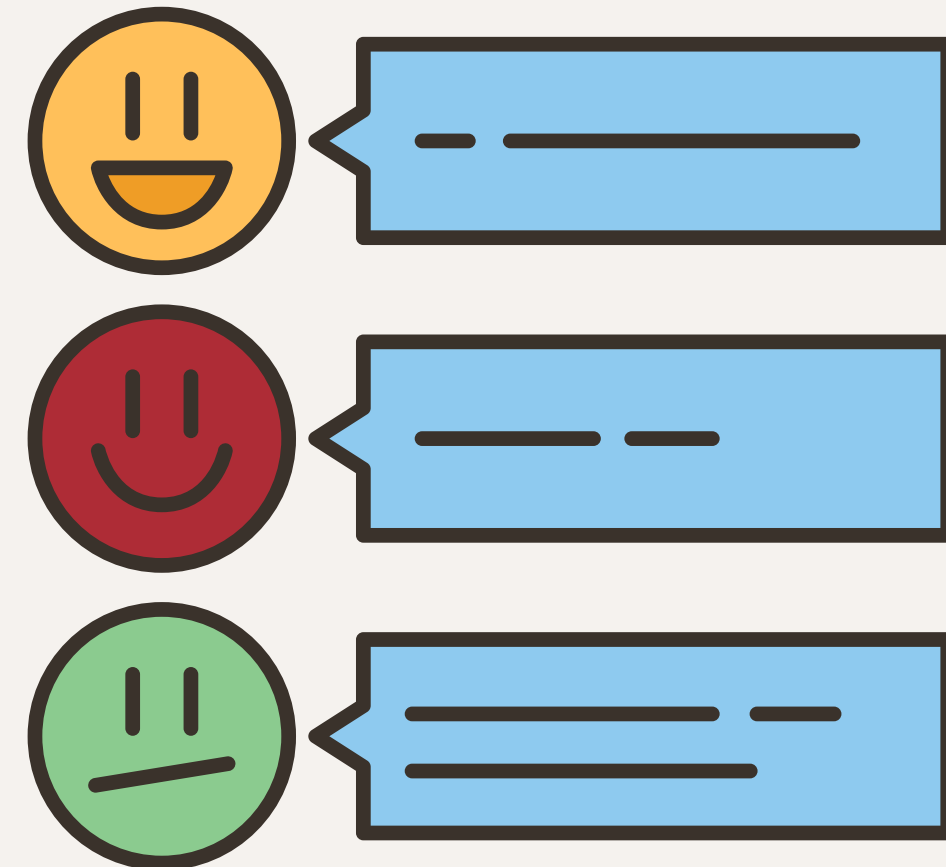
Selected columns: reviews.text, reviews.rating

Mapped star ratings:

1-2 ★ → Negative (0)

3 ★ → Neutral (1)

4-5 ★ → Positive (2)



Model Selection

- **Model**

DistilBERT (Transformer-based model)



- **Why**

- Lightweight version of BERT, pretrained on large corpus, suitable for fine-tuning on text tasks

Training and Validation

- Data Split: 80% Training, 20% Validation
- Epochs: 4
- Optimizer: AdamW
- Loss Function: CrossEntropyLoss
- Batch Size: 16

Results

- Accuracy: 93.0%
- Precision: 94.5%
- Recall: 93.0%
- F1 Score: 93.7%

DISTILBERT EVALUATION

[3752/3752 11:48, Epoch 4/4]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	0.565400	0.376724	0.889359	0.939206	0.889359	0.907807
2	0.156600	0.385973	0.913887	0.945544	0.913887	0.925540
3	0.067600	0.350771	0.933475	0.948533	0.933475	0.939368
4	0.041200	0.402585	0.926769	0.949377	0.926769	0.935356

[355/355 00:20]

{'eval_loss': 0.35077106952667236,

'eval_accuracy': 0.9334745014999117,

'eval_precision': 0.9485326192925891,

'eval_recall': 0.9334745014999117,

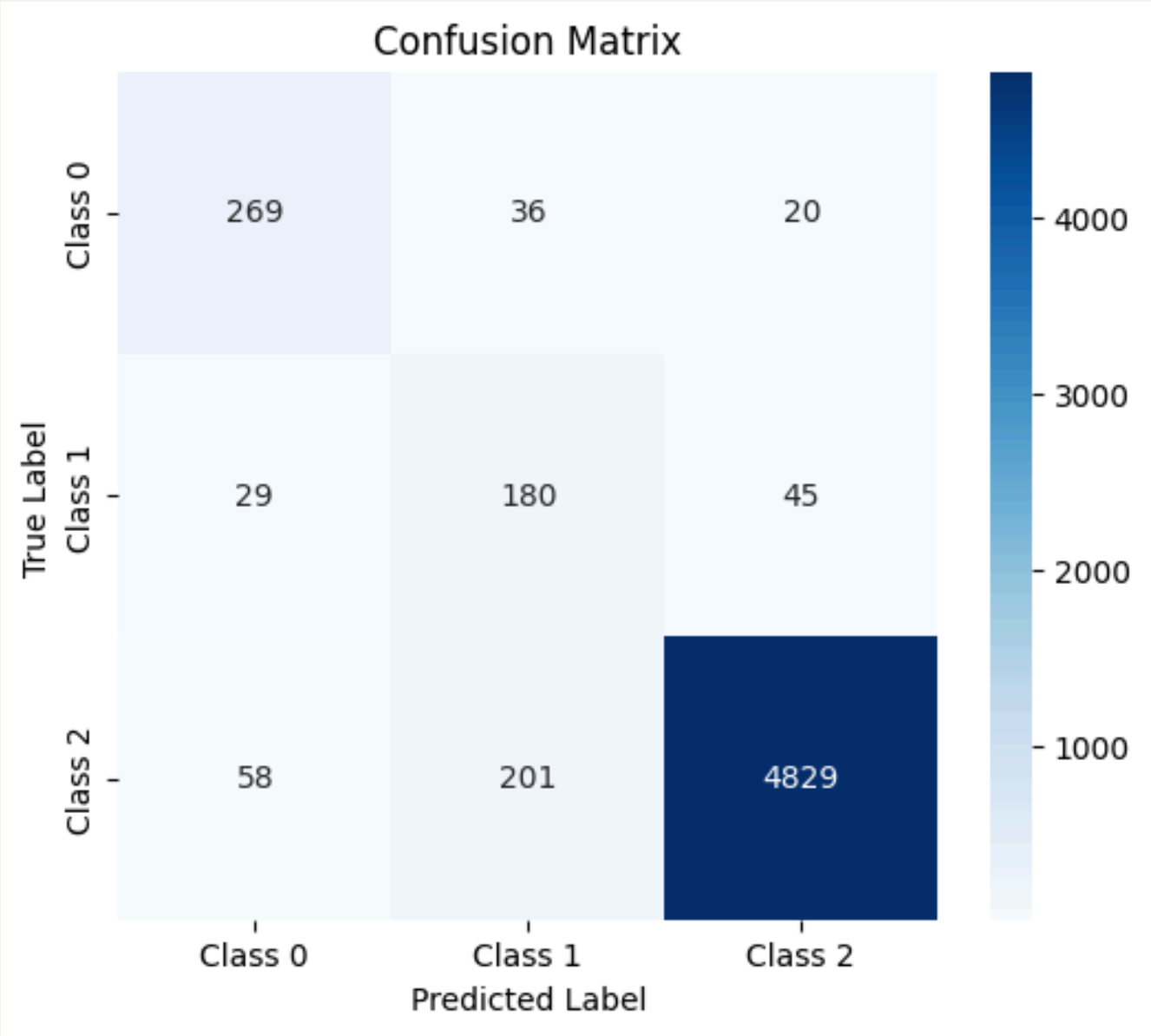
'eval_f1': 0.9393678744643686,

'eval_runtime': 20.7935,

'eval_samples_per_second': 272.537,

'eval_steps_per_second': 17.073,

'epoch': 4.0}



02

PRODUCT
CATEGORY
CLUSTERING

PRODUCT CATEGORY CLUSTERING

GROUP THOUSANDS OF PRODUCTS INTO A FEW MEANINGFUL META-CATEGORIES TO SIMPLIFY DOWNSTREAM ANALYSIS AND RECOMMENDATIONS

Preprocessing the Data

Combined categories + primaryCategories into a unified text field.
Cleaned and normalized text (removed special characters, lowercased)

Embedding

Used Sentence-BERT (allMiniLM-L6-v2) to convert category text into vector representations.

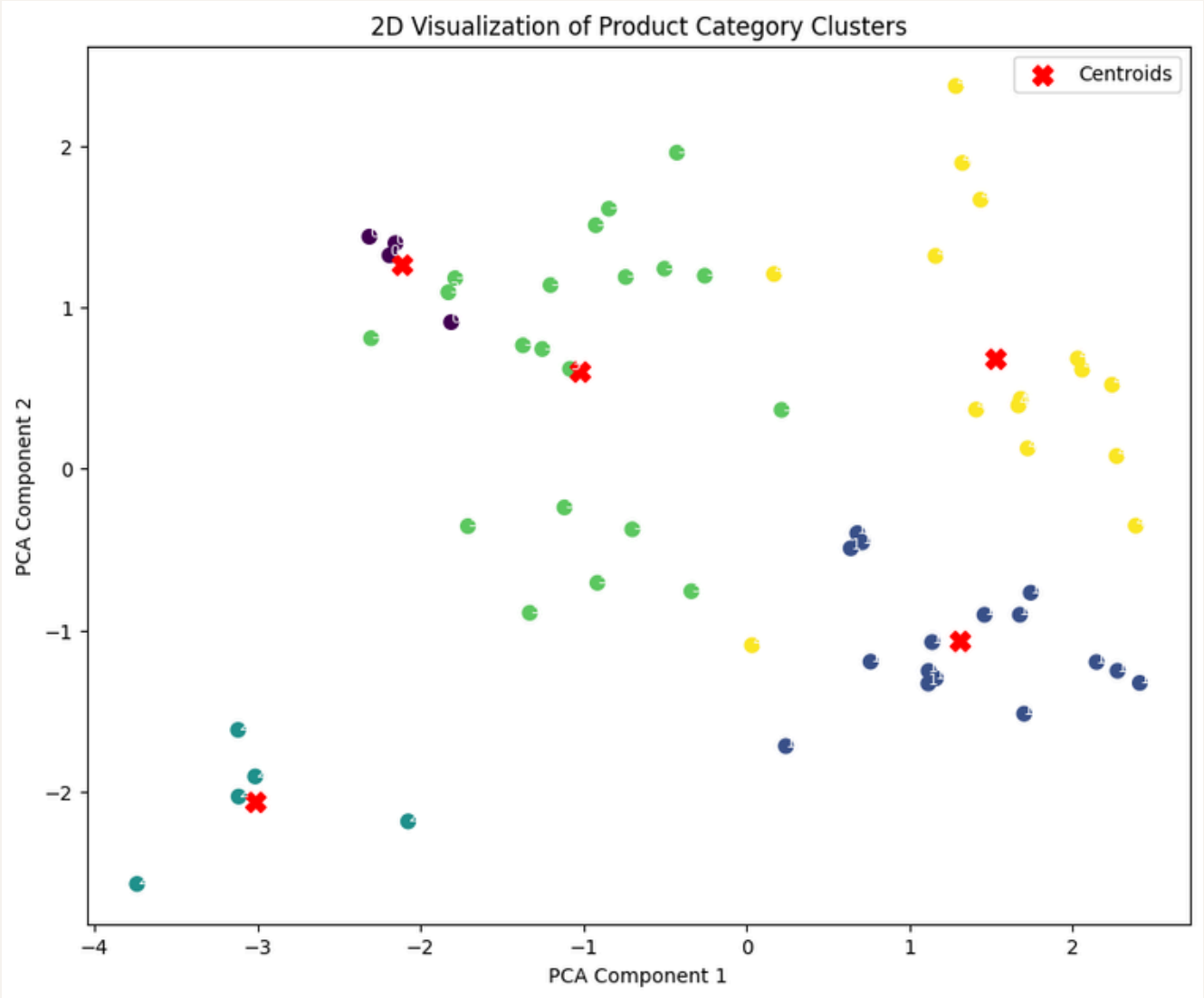
Clustering

Applied K-Means clustering on embeddings (n_clusters = 5).
Mapped clusters to interpretable metacategories .

```
cluster_names = {
    0: "Pet Supplies & Small Home Goods",
    1: "Fire Tablets & Accessories",
    2: "Amazon Echo & Smart Speakers",
    3: "AmazonBasics & General Tech Accessories",
    4: "Kindle E-Readers & Charging Accessories"
}
```

12835	all new kindle e reader black glare free touch...	4	Kindle E-Readers & Charging Accessories
12852	amazonbasics usb cable a male to b male feet m...	3	AmazonBasics & General Tech Accessories
12858	two door top load pet kennel travel crate dog ...	0	Pet Supplies & Small Home Goods
12859	oem amazon kindle power usb adapter wall trave...	4	Kindle E-Readers & Charging Accessories
12863	amazonbasics gauge speaker wire feet	3	AmazonBasics & General Tech Accessories
12868	amazon echo nd generation smart assistant oak ...	3	AmazonBasics & General Tech Accessories
12871	kindle voyage e reader high resolution display...	4	Kindle E-Readers & Charging Accessories
13376	amazonbasics nespresso pod storage drawer caps...	3	AmazonBasics & General Tech Accessories
13377	certified refurbished amazon echo	2	Amazon Echo & Smart Speakers
13380	all new echo nd generation with improved sound...	4	Kindle E-Readers & Charging Accessories
13386	all new fire hd tablet hd display wi fi gb inc...	1	Fire Tablets & Accessories
13387	fire kids edition tablet display wi fi gb pink...	1	Fire Tablets & Accessories
13404	all new fire hd tablet hd display wi fi gb inc...	1	Fire Tablets & Accessories
17687	kindle oasis e reader with leather charging co...	4	Kindle E-Readers & Charging Accessories
17688	kindle oasis e reader high resolution display...	4	Kindle E-Readers & Charging Accessories

VISUALIZATION



03

REVIEW

SUMMARIZATION WITH
GENERATIVE AI

REVIEW SUMMARIZATION WITH GENERATIVE AI

GENERATE INSIGHTFUL SUMMARIES OF CUSTOMER REVIEWS FOR TOP-PERFORMING AND POOR-PERFORMING PRODUCTS IN EACH META-CATEGORY.

Model

Applied OpenAI's gpt-3.5-turbo.

When it comes to Amazon Echo & Smart Speakers, having the right accessories can make a world of difference in enhancing your overall experience. Let's take a look at the top 3 recommended products in this category and also highlight one product you might want to steer clear of.

****Top 3 Recommended Products:****

1. ****AmazonBasics Gauge Speaker Wire (100 Feet):**** This speaker wire is perfect for setting up your audio system with ease. It provides clear sound transmission and is durable for long-term use.
2. ****AmazonBasics USB Cable A Male to B Male (6 Feet/1.8 Meters):**** This USB cable is essential for connecting your smart devices and peripherals. With a convenient length of 6 feet, it offers flexibility and fast data transfer.
3. ****Expanding Accordion File Folder:**** Stay organized with this portable document organizer. Ideal for storing letters and important papers, this file folder is a must-have for keeping your workspace tidy.

****Key Differences:****

- The speaker wire is designed for audio setups, while the USB cable is for connecting devices.
- The USB cable offers fast data transfer, whereas the file folder focuses on organization.
- Each product serves a different purpose but all aim to improve functionality and convenience.

****Top Complaints:****

- ***AmazonBasics Gauge Speaker Wire (100 Feet):*** No major complaints reported.
- ***AmazonBasics USB Cable A Male to B Male (6 Feet/1.8 Meters):*** No significant complaints mentioned.
- ***Expanding Accordion File Folder:*** No notable complaints highlighted.

****Product to Avoid:****

- ****Amazon Kindle Replacement Power Adapter:**** Despite being a necessary accessory for Kindle users, this power adapter has garnered negative feedback. Users have complained about the poor quality of the cord, with reports of it cracking and falling apart after minimal use. Some users have expressed frustration over the adapter failing to work properly, leading to inconvenience and dissatisfaction.

In conclusion, when looking for accessories for your Amazon Echo & Smart Speakers, be sure to choose reliable products like the AmazonBasics speaker wire, USB cable, or file folder to enhance your setup and organization. And remember to steer clear of the Amazon Kindle Replacement Power Adapter to avoid potential issues with durability and functionality. Make informed choices to elevate your smart speaker experience!

CONCLUSION

This project shows how NLP can transform raw review data into actionable business insights from classification to summarization. It provides companies with a scalable way to understand customer needs and market products more effectively.